

FUNDAMENTALS OF NANO-ELECTRONICS

Basic Concepts

The New Perspective



2. Energy Band Model

3. What and Where

is the Voltage?

4. Heat & Electricity:

Second Law & Information

1.1. Introduction

1.2. Two Key Concepts

1.3. Why Electrons Flow

1.4. Conductance Formula

1.5. Ballistic(B) Conductance

1.6. Diffusive(D) Conductance

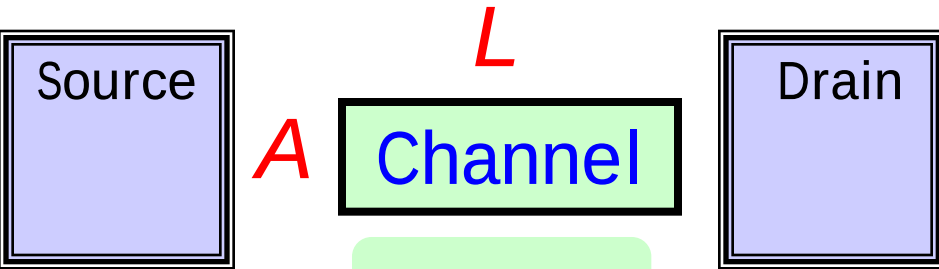
1.7. Connecting B to D

1.8. Angular Averaging

1.9. Drude Formula

1.10. Summing up ..

1.6a Diffusive Conductance



Ballistic

$$t_B = \frac{L}{\bar{v}}$$

$$D \sim AL$$

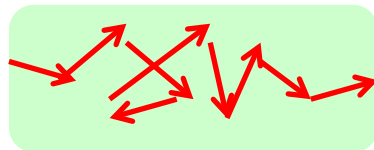
$$G_B = \frac{q^2 D \bar{v}}{2L} \sim A$$

$$G = \frac{q^2 D}{2t}$$

Diffusive

$$t_D = \frac{L^2}{2\bar{D}}$$

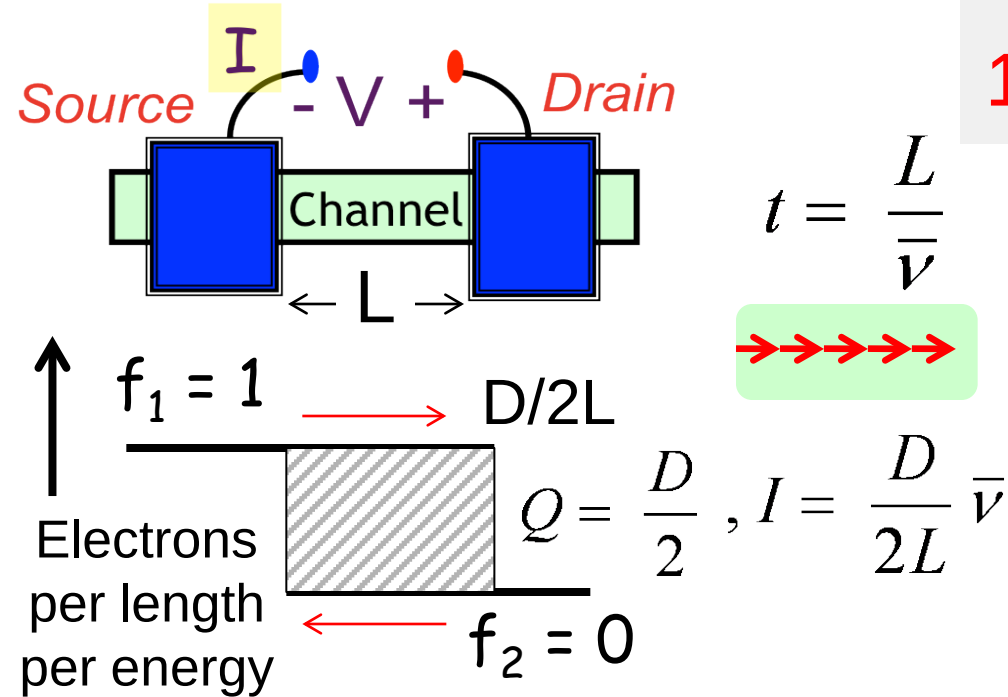
$$G = \frac{A}{L} \overbrace{q^2 \frac{D}{AL} \bar{D}}^{\sigma}$$



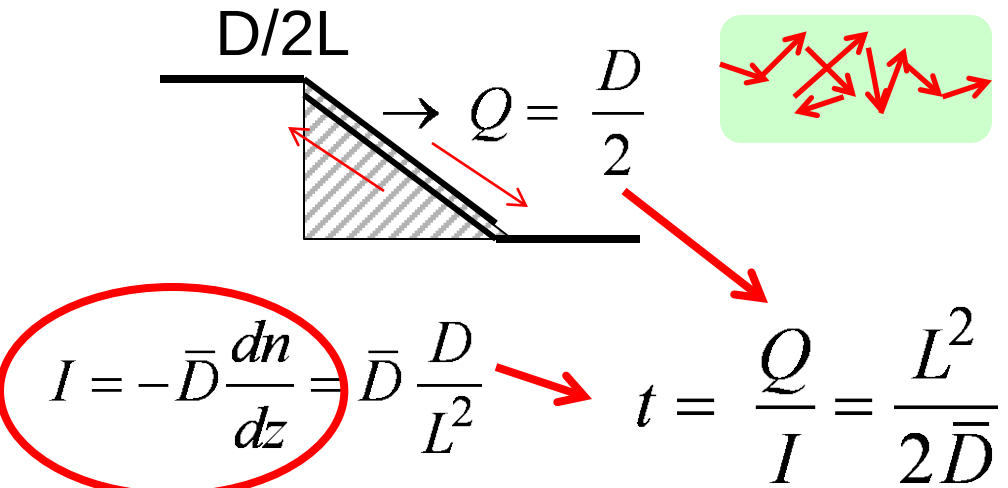
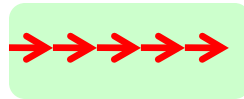
Drude
formula

$$\sigma = \frac{q^2 n \tau}{m}$$

1.6b Diffusive Conductance



$$t = \frac{L}{\bar{v}}$$



$$Q = I \times t$$

of electrons in channel Electrons /second Time in channel

of PhD students PhD's /year Time to PhD

$$t_D = \frac{L^2}{2\bar{D}} \quad t_B = \frac{L}{\bar{v}}$$

$$\bar{D} = \langle v_z^2 \tau \rangle \quad G = \frac{q^2 D}{2t}$$

Berg: Random Walks in Biology, Princeton 1993

$$I/V = \int_{-\infty}^{+\infty} dE \left(-\frac{\partial f_0}{\partial E} \right) G(E)$$

$$G(E) = \frac{q^2 D}{2t} = \frac{G_B \lambda}{L + \lambda}, \quad \lambda \equiv \frac{2\bar{D}}{\bar{v}}$$

$$\sigma A = G_B \lambda$$

$$\bar{v} = \langle |v_z| \rangle, \quad \bar{D} = \langle v_z^2 \tau \rangle$$

Ballistic $G_B = q^2 \frac{D\bar{v}}{2L}$

Diffusive $\sigma = q^2 \bar{D} \frac{D}{AL}$

Coming up next ..

1.1. Introduction

1.2. Two Key Concepts

1.3. Why Electrons Flow

1.4. Conductance Formula

1.5. Ballistic(B) Conductance

1.6. Diffusive(D) Conductance

1.7. Connecting B to D

1.8. Angular Averaging

1.9. Drude Formula

1.10. Summing up ..